

Homework 6

ORF 245 - Fundamentals of Statistics

Due: Apr. 4th, 2025, End-of-Day (11:59pm)

Notes:

- Collaboration with other students is encouraged, but you must i) write up your own answers, and ii) acknowledge any sources or students you consulted.
 - Submit the homework on Gradescope, and be sure to assign each page to the relevant question(s).
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Exercise: 1. (Bias-Variance Decomposition)

Let $\hat{\theta}(X_1, \dots, X_n)$ be any point estimator for some unknown population parameter θ . Recall that the Mean Squared Error (MSE) for an estimator $\hat{\theta}(\cdot)$ is defined as:

$$\text{MSE}(\hat{\theta}; \theta) = \mathbb{E} \left[(\hat{\theta} - \theta)^2 \right].$$

Show that $\text{MSE}(\hat{\theta}; \theta) = \text{Bias}(\hat{\theta}; \theta)^2 + \text{Var}(\hat{\theta})$.

Exercise: 2. (MLE of Poisson)

Let the discrete random variable X has a Poisson distribution with unknown parameter $\lambda > 0$. The pmf of X is given by:

$$p(x; \lambda) = \frac{e^{-\lambda} \lambda^x}{x!} \quad \text{for } x = 0, 1, 2, \dots$$

Recall that a Poisson random variable appropriately models the number of arrivals within a given unit of time, and that its mean is given by λ .

- (a) If we observe 8 arrivals in one unit of time, compute the likelihood and log-likelihood functions. (I.e. we obtain a single realization of X given by $x = 8$.)

Note: the log likelihood is defined as the (natural) logarithm of the likelihood function.

- (b) Find the maximum likelihood estimate of λ given the setting in part (a), where we observe 8 arrivals, i.e. find $\hat{\lambda}_{MLE}(8)$.

- (c) Now, suppose that we have $X_1, \dots, X_n \stackrel{i.i.d.}{\sim} \text{Poi}(\lambda)$. Find the likelihood and log-likelihood functions.

- (d) Find the maximum likelihood estimator $\hat{\lambda}_{MLE}(X_1, \dots, X_n)$ for λ .
- (e) Is the estimator $\hat{\lambda}_{MLE}(X_1, \dots, X_n)$ biased? Justify your answer.
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Exercise: 3. (MLE of a New Distribution)

Let $X_1, \dots, X_n$ be i.i.d. random variables, with each of their marginal pdfs given by:

$$f(x; \nu) = \begin{cases} 0, & \text{if } x < 0 \\ \nu^{-x} \ln(\nu), & \text{if } x \geq 0, \end{cases} \quad (1)$$

where $\nu > 1$ is an unknown parameter.

- (a) Compute the likelihood function $\mathcal{L}(\nu; X_1, \dots, X_n)$.
- (b) Find the maximum likelihood estimator $\hat{\nu}_{MLE}(X_1, \dots, X_n)$.
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Exercise: 4. (MLE for Variance of a Normally Distributed Population)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \theta)$ be i.i.d. random variables from a Normal distribution with mean μ and variance θ .

In this problem, **assume the population mean μ is known**, and the population variance θ is unknown.

- (a) Compute the likelihood function $\mathcal{L}(\theta; X_1, \dots, X_n)$.
- (b) Compute the maximum likelihood estimator $\hat{\theta}_{MLE}(X_1, \dots, X_n)$ for the population variance θ .
- (c) Is $\hat{\theta}_{MLE}(X_1, \dots, X_n)$ an unbiased estimator? Justify your answer. (A proof is not necessary, but certainly acceptable.)
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Exercise: 5. [OPTIONAL] (Point Estimation for Uniform RV's)

Note: This is an *optional* exercise – it will not be graded, will not factor into your grade (either positively or negatively), but we will release solutions after the submission deadline.

Let $X_1, \dots, X_n$ be i.i.d. random variables according to a uniform distribution on the interval $[0, \theta]$, where $\theta > 0$ is an unknown parameter. Each of their marginal probability density function is given by:

$$f(x; \theta) = \begin{cases} \frac{1}{\theta} & \text{if } x \in [0, \theta] \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Show that $\hat{\theta}_{AVG}(X_1, \dots, X_n) = 2\bar{X}_n$ is an unbiased estimator of θ .
- (b) Compute $\text{Var}(\hat{\theta}_{AVG}(X_1, \dots, X_n))$.

- (c) Let $X_{(n)} = \max_{i=1 \dots n} X_i$ denote the largest of all X_i 's. Show that the cumulative distribution function of $X_{(n)}$ is given by:

$$F_{(n)}(x; \theta) = \begin{cases} 0 & \text{if } x < 0 \\ \left(\frac{x}{\theta}\right)^n & \text{if } x \in [0, \theta] \\ 1 & \text{if } x > \theta. \end{cases}$$

- (d) Show that the maximum likelihood estimator for θ is given by $\hat{\theta}_{MLE}(X_1, \dots, X_n) = X_{(n)}$.

Hint: In this particular example, the likelihood function will not be differentiable for all values of θ – and so you won't be able to use our usual approach to find its argmax. Instead, study the likelihood function itself – where is it equal to zero and why? Where is it not differentiable?

- (e) Compute $\mathbb{E}[\hat{\theta}_{MLE}(X_1, \dots, X_n)]$. Is this estimator unbiased? If not, how could you construct a different unbiased estimator from $\hat{\theta}_{MLE}(X_1, \dots, X_n)$?
- (f) Compute $\text{Var}(\hat{\theta}_{MLE}(X_1, \dots, X_n))$.